

Performance Testing

Date	14 July 2026
Team ID	
Project Name	SMART LENDER
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	/predict API Endpoint
Test Environment	Localhost development server
Test Date	14 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent user load test	100	60	Response time < 1 second
2	Ramp-up spike test	500	30	No server crash / low errors
3	Stress test	1000	120	Graceful handling under heavy load
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.15 seconds	Pass	Excellent responsiveness
2	Response Time (Max)	< 5 seconds	0.85 seconds	Pass	Within acceptable bounds
3	Throughput (Req/sec)		45 req/sec	Pass	Adequate for bank staff usage
4	Error Rate	< 1%	0.0%	Pass	Reliable operation
5	CPU Utilization	< 80%	15%	Pass	Low CPU usage
6	Memory Utilization	< 80%	45%	Pass	No memory leaks observed

Step 4: Observations & Analysis

- Response Times:** The average response time was 0.15 seconds, and the peak response time was 0.85 seconds. Both are well below the target limits (< 2.0s and < 5.0s respectively), ensuring an extremely fast user experience.
- Throughput Capacity:** The system maintained a steady throughput of 45 requests/second, which is highly sufficient for handling concurrent loan queries from multiple bank staff members.
- System Reliability:** An error rate of 0.0% was achieved, demonstrating high stability and fault tolerance of the Flask backend and XGBoost model execution under load.
- Resource Footprint:** CPU utilization remained low at 15% and memory usage was stable at 45%. No memory leaks or resource exhaustion were observed, verifying the efficiency of the application implementation.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

